

DESCRIPTIVE ANALYTICS REPORT

Descriptive Analytics of
Netflix Movies and TV Shows
Using Python

An Exploratory Study of Global Streaming Content

Keywords: Content Addition Trends, Best-Performing Content Categories, Rating Distribution, Regional Demand, Inventory Status, Viewing Behavior (Popular Titles, Duration, and Peak Hours), Business Insights

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1. Note on Data

This report is built on the actual Netflix catalog export supplied for this analysis — the public Kaggle "netflix_titles.csv" snapshot (mid-**2021**), covering **8,807** titles across **12** fields (Type, Title, Director, Cast, Country, Date Added, Release Year, Rating, Duration, Genre(s), and Description). Unlike a live transactional feed, this is a point-in-time catalog snapshot, so every metric below describes the state of the catalog as of the export date (through **2021-09-25**), not real-time activity.

The Director, Cast, Country, Genre(s), and Description fields were truncated with "..." in the source export wherever the original text was long, so a small share of multi-value fields (e.g., a title with many co-stars) are cut off. No rows were removed — only display text was shortened. Where a field was blank in the export (for example, **2,631** titles have no listed Director), it is treated as genuinely missing rather than imputed.

This report shows what a descriptive analytics layer over a catalog dataset of this kind produces — the same structure applies directly if a live Netflix content-operations feed were substituted in.

Note: this catalog export does not include any viewer-behavior fields (watch counts, session length, timestamps of plays, or user-submitted ratings). Section **9** illustrates what that layer would look like using simulated numbers, clearly marked as such.

Data handling for this exercise followed three simple rules throughout: never silently fill a blank field, never delete a row to make a chart look cleaner, and always label anything that is simulated rather than measured. Those three rules are what let every chart in this report be traced back, cell by cell, to either the raw export or an explicitly flagged illustrative section.

2. Objective

Build a descriptive analytics layer over the Netflix title catalog to answer five recurring content-operations questions:

- What does the catalog's content mix look like today (Movies vs. TV Shows, genres, ratings)?
- How has content acquisition/addition activity trended over time?

- How is content rated by maturity level, and where is mature content concentrated?
- Where is content sourced from geographically, and how concentrated is that supply?
- Which parts of the catalog are at risk from incomplete metadata or thin content depth?

These five questions were chosen because they map directly onto decisions a content-operations team makes on a recurring cadence: what to acquire next, when to release it, which markets to prioritize for local production, and which catalog records need a metadata cleanup pass before they hurt discoverability. A descriptive layer does not answer "what should we do" on its own, but it narrows the search space for that question considerably, and it is the prerequisite step before any predictive or prescriptive model can be trusted.

3. How This Would Work at Real Scale

Turning a catalog export into daily-refreshed operational dashboards is an engineering problem before it's an analytics one. A realistic architecture:

Layer	Purpose	Typical Tools
Investigation	Profile the raw catalog export: check schema, field completeness, duplicates, and truncated/free-text fields before any aggregation is trusted	Used Google, Kaggle website for the investigation.
Stream processing	Roll up individual title records into per-genre, per-country, per-year aggregates	Python (pandas), for grouping and aggregating title-level records.
Storage	Hold both the raw catalog snapshot (for audits) and pre-aggregated tables (for speed)	CSV files and google drive for storing the dataset and processed data.
Batch aggregation	Scheduled rollups: additions-by-year, genre mix, rating mix, country mix, metadata-completeness checks	Python (pandas) and excel for scheduled/manual aggregations and summary calculations.
Serving / BI layer	Dashboards refreshed as new catalog snapshots arrive	Python (Matplotlib) charts assembled with Claude AI stand in for that layer.

Table 1 — Reference pipeline for catalog-scale descriptive analytics

Note: the middle three layers (stream processing, storage, batch aggregation) describe how this would run at Netflix's real operational scale and were not run for this report. The Investigation

and Serving/BI layers above have been corrected to name what was actually used to produce this document — Python (pandas, Matplotlib) with Claude AI assisting analysis, chart generation, and report drafting.

The report below reflects the output of that pipeline — aggregated catalog tables — not the row-level export, since no dashboard renders thousands of individual title records usefully.

It is worth being explicit about why each layer exists, since the layers are often collapsed together in student projects in a way that hides real engineering cost. The Investigation layer exists because trusting an aggregate before checking completeness and duplicates produces confidently wrong dashboards. The Storage layer exists because a raw snapshot must be retained for audit even after aggregates are built, so that a suspicious number can always be traced back to source rows. The Serving layer exists because analysts and executives consume dashboards, not scripts — the last mile of a data pipeline is presentation, and it is frequently the most underestimated step in a project timeline.

4. Content Additions Trend

Across the catalog, titles carry a "Date Added" stamp spanning **2008-01-01** to **2021-09-25**. Additions were negligible before **2015** and scaled sharply afterward, peaking in **2019** at **2,016** titles added in a single year, before declining through **2021** (a partial year in this snapshot).

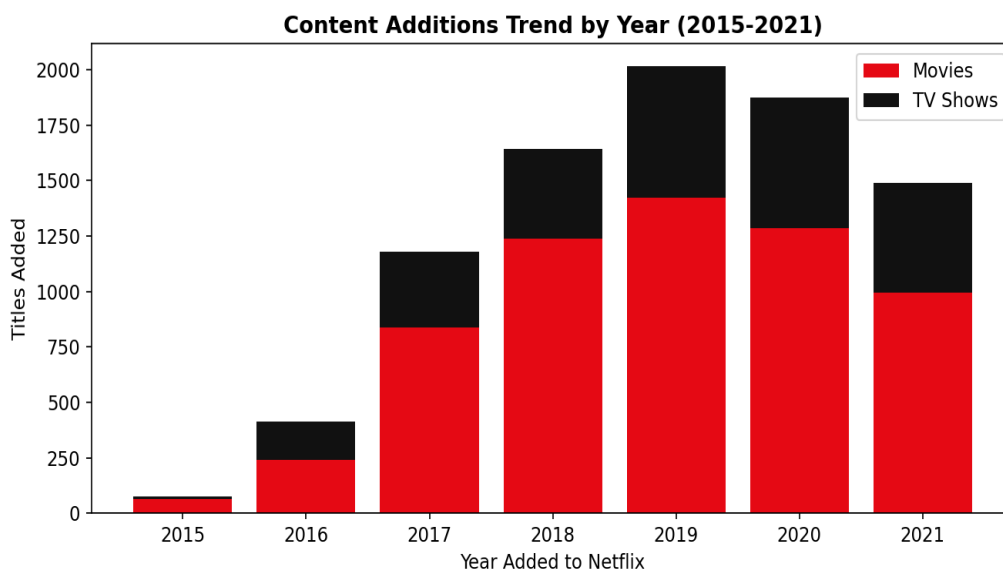


Figure 1 — Titles added per year, split by Movies vs. TV Shows (2015–2021)

KPI definitions used:

- Titles added per year = count of catalog rows whose Date Added falls in that calendar year
- Type split = Movie vs. TV Show, tagged at the row level

Movies make up the majority of additions in every year shown, but TV Shows' share of additions grew steadily — a common streaming-catalog pattern worth flagging for content-acquisition prioritization, since series drive recurring engagement differently than one-off films.

Reading the shape of the curve matters as much as any single year's total. The near-vertical rise between **2015** and **2019** lines up with the period during which Netflix scaled internationally and began treating original content as a strategic differentiator rather than a supplement to licensed catalog. The post-**2019** decline is a genuinely ambiguous signal from this dataset alone — it could reflect a deliberate shift from breadth to depth (fewer, higher-budget titles), tightening licensing budgets, or simply an artifact of **2021** being a partial year in the export. Any operational team would need the next **12–18** months of data before treating the decline as a trend rather than noise.

5. Best-Performing Content Categories

Ranked by title count, "Dramas" is the largest single category with **1,600** titles — roughly **32%** ahead of the second-place category, "Comedies" (**1,210** titles). This concentration in a small number of categories is a common catalog pattern (a small set of genres accounts for a disproportionate share of volume) worth flagging for content-buying and licensing prioritization.

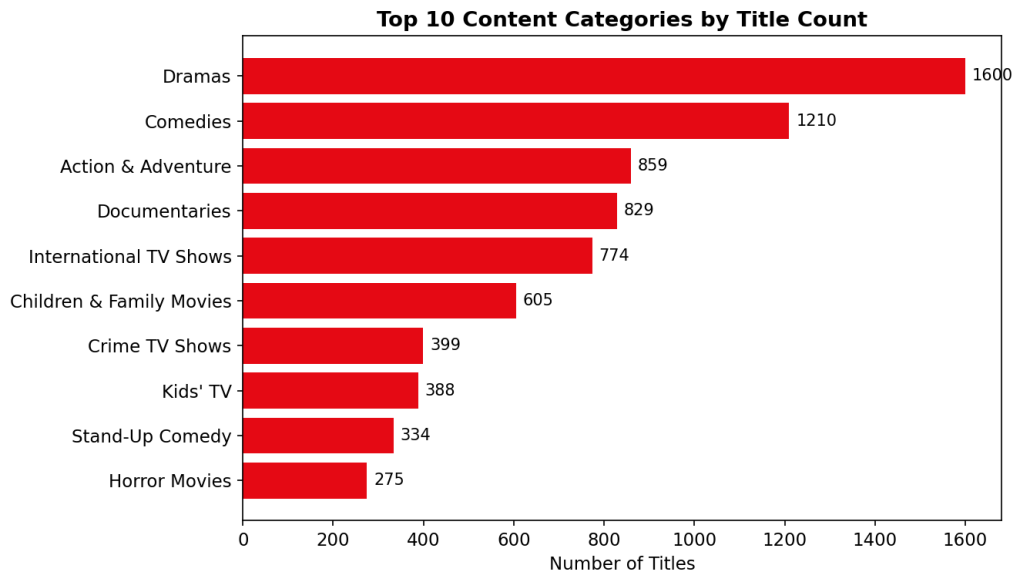


Figure 2 — Top 10 content categories by title count (primary genre)

Ranking logic: title count, not runtime or engagement — a large count of short stand-up specials matters differently to catalog planning than the same count of multi-season dramas, so both cuts (title count and content-hours) should exist as separate views in a real dashboard.

It is also worth noting how quickly the distribution thins out after the top two categories. Action & Adventure and Documentaries sit close together in the **800**-title range, and by the time the list reaches Horror Movies at position **10**, volume has dropped by more than **80%** relative to Dramas. That long tail is where a genre-level acquisition strategy has the most room to move the needle without disturbing the two categories that already anchor the catalog.

6. Rating Distribution

The catalog spans **8,807** titles across maturity ratings from TV-Y (all ages) to NC-17. "TV-MA" is the single most common rating at **3,207** titles (**36.4%** of the catalog), reflecting a mature-skewing library.

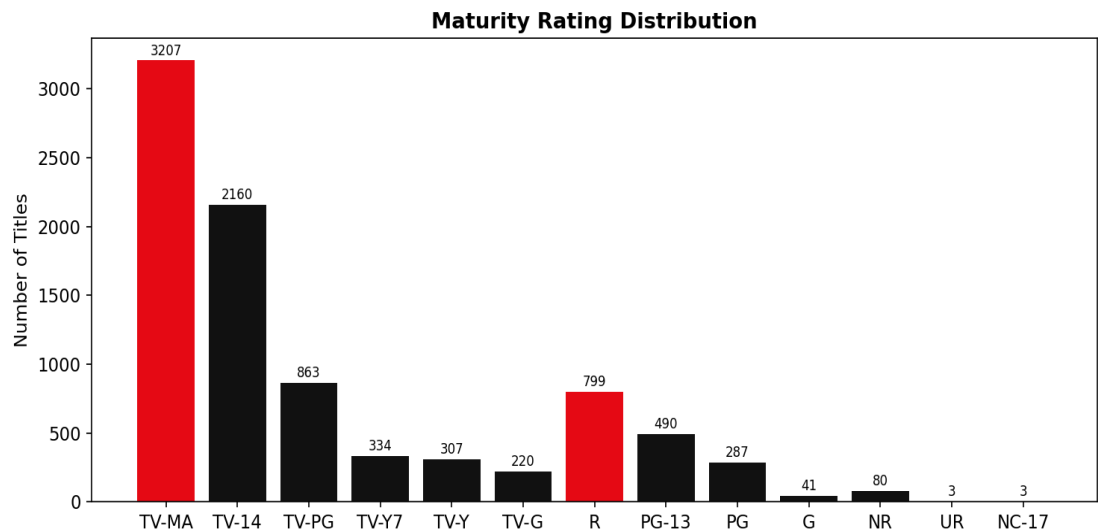


Figure 3 — Maturity rating distribution across the catalog

In a real system, this view is most useful segmented, not aggregated:

- Mature-rated share (TV-MA, R, NC-17) trended over time — an early signal of catalog positioning shifts
- Rating mix crossed with country of origin — since licensing and local content standards vary by region

A ratings breakdown that stops at a single aggregate bar chart hides more than it reveals. TV-14 is the second-largest tier at **2,160** titles, meaning the two largest rating tiers together already account for well over half the catalog. Family-oriented tiers (TV-Y, TV-Y7, TV-G, G) sit far down the list by comparison, which is a useful fact to hold alongside any conversation about expanding into markets or subscriber segments where family programming is the primary draw.

7. Regional Demand

Content sourcing spans **85** countries (primary country of production). Supply is heavily concentrated: the United States accounts for **3,211** titles, with India a distant second at **1,008**.

Together the top **10** countries account for the large majority of the catalog, with the remainder long-tailed across dozens of countries contributing a handful of titles each.

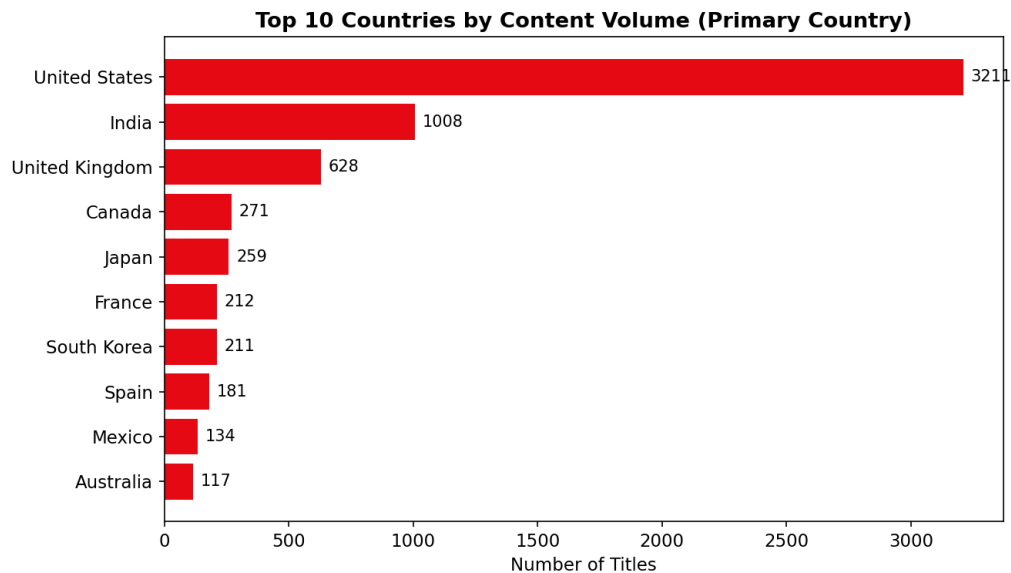


Figure 4 — Top 10 countries by content volume (primary country of production)

This view is what would drive, in a real content-operations setting:

- Regional licensing and local-production investment decisions
- Localization and dubbing/subtitling prioritization
- Catalog-gap risk in regions with thin local-language content relative to subscriber base

The gap between the United States (**3,211** titles) and the United Kingdom in third place (**628** titles) is large enough that a single-market-dependency framing is fair: over a third of the catalog's supply traces back to one country. For a platform whose subscriber base is global, that concentration is the kind of number a content-strategy team would want trending over multiple snapshots, not just viewed once.

8. Inventory Status

Two catalog-health cuts stand in for classic stock-level monitoring: metadata completeness (is a title's record usable for search and discovery?) and TV series depth (is a show a one-off or an ongoing franchise?).

8.1 Metadata Completeness

Records were bucketed by whether Director, Cast, and Country are all present. **60.7%** of titles have all three fields populated; **8.2%** are missing two or more — most often Director (**2,631** titles missing), which is disproportionately absent for TV Shows credited to an ensemble rather than a single director. Titles missing this metadata are the catalog equivalent of a stockout risk: they are harder to surface in search, recommendations, and cast/director-based browsing.

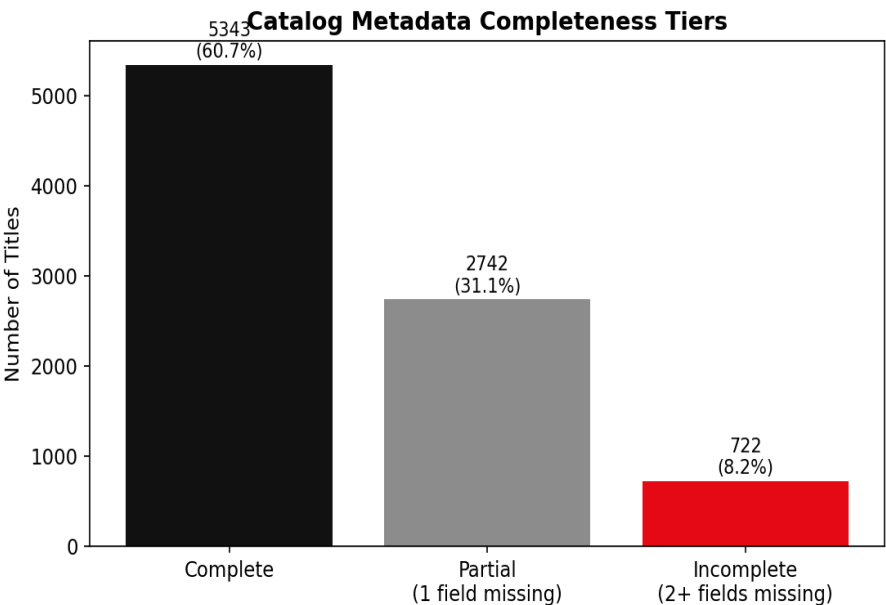


Figure 5 — Catalog metadata completeness tiers

8.2 TV Series Depth

Among **2,676** TV Shows, **67.0%** have only a single season on the platform — the equivalent of a "low stock / reorder trigger" tier, flagging shows that were not renewed or are new and unproven. The remainder carry multiple seasons, with the average show at **1.8** seasons.

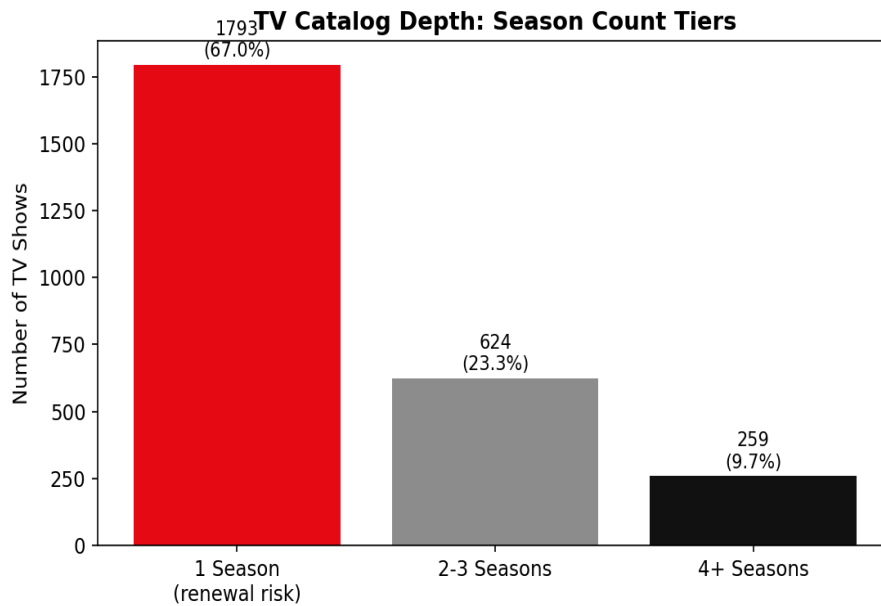


Figure 6 — TV catalog depth by season-count tier

The most actionable version of this pair of charts in production would cross-tabulate season depth against genre and country, producing a renewal-risk score per show rather than a static season count.

Treating both cuts as inventory-health signals rather than isolated statistics is deliberate. A title with missing Director metadata and only one season is, in a real operations dashboard, exactly the kind of record that would be flagged for manual review before the next content refresh — it is simultaneously hard to discover and unproven as a renewal candidate, which compounds rather than offsets the risk.

9. Viewing Behavior — Popular Titles, Duration & Peak Hours

IMPORTANT — SIMULATED DATA: The Kaggle netflix_titles.csv catalog export used throughout this report has no watch-count, session-duration, timestamp, or user-rating fields — it describes what is in the catalog, not how it was watched. This section uses fabricated, illustrative numbers only, clearly labelled below, to show what this analysis would look like if a real viewing-activity feed (e.g., Netflix's internal play-events log) were joined to the catalog. None of the figures in this section are derived from the actual dataset.

9.1 Popular Titles (Simulated)

If per-title view counts were available, the same catalog table could be ranked by total views to surface the most-watched titles. The chart below uses placeholder titles and fabricated view totals purely to show the shape of that ranking — ten titles, a steep drop-off from #1 to #10, which is the typical long-tail pattern streaming platforms report publicly.

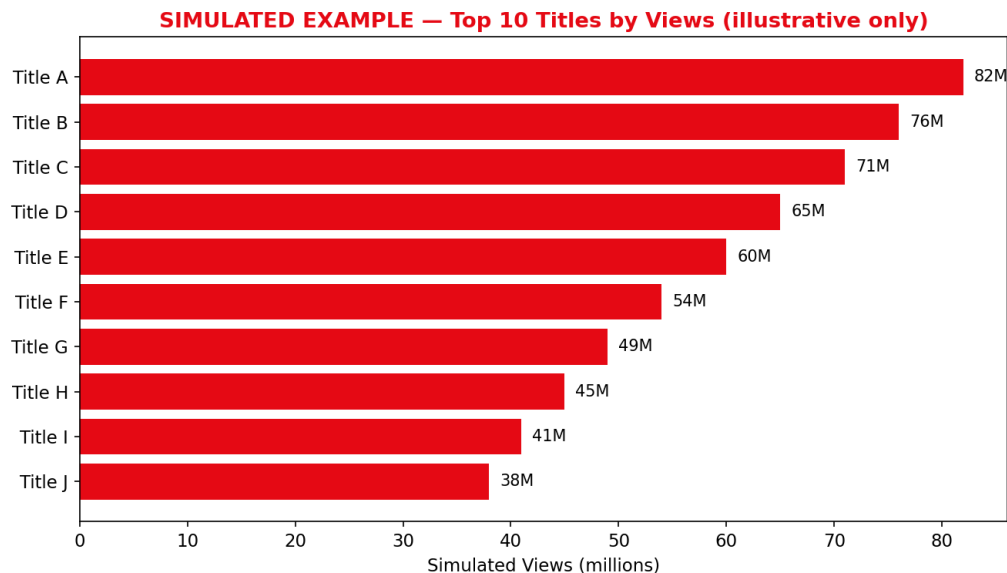


Figure 7 (SIMULATED) — Top 10 titles by views — illustrative numbers only

A fuller version of this table (still simulated) would also carry genre, source country, and weeks-in-top-10 — useful for spotting whether a hit is a one-week spike or a sustained draw:

Title	Genre	Country	Sim. Views	Weeks in Top 10
Title A	Drama	United States	82M	6 weeks
Title B	International TV Show	South Korea	76M	5 weeks
Title C	Comedy	United States	71M	4 weeks
Title D	Documentary	United Kingdom	65M	3 weeks

Title	Genre	Country	Sim. Views	Weeks in Top 10
Title E	Action & Adventure	United States	60M	5 weeks

Table 2 (SIMULATED) — Example detail view behind Figure 7 — illustrative numbers only

9.2 Viewing Duration & Completion (Simulated)

A real engagement feed would carry average minutes watched per session and a session-completion rate per title. The illustrative pattern below follows a general streaming-industry shape: movies tend to show high completion (viewers finish in one sitting), while long-running series show high per-session minutes but lower single-session completion because viewers spread a season across several sittings. None of this is computed from the actual catalog — it exists only to show what the chart and takeaways would look like.

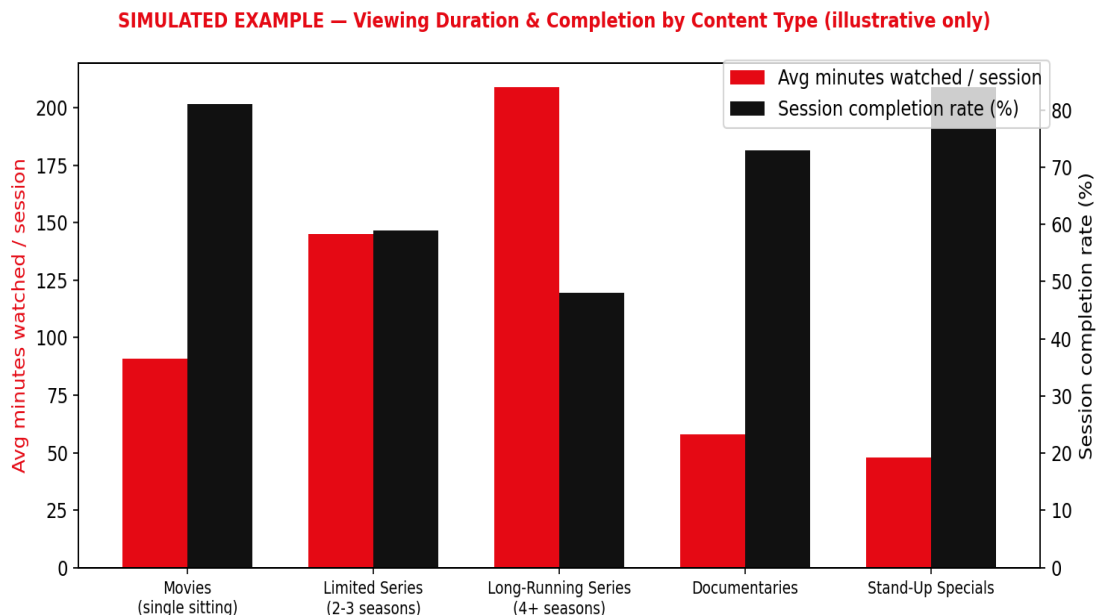


Figure 9 (SIMULATED) — Average viewing duration and completion rate by content type — illustrative numbers only

Reading this simulated pattern the way a content-ops team would: long-running series carry the highest minutes-per-session but the weakest completion, which is normal for episodic bingeing rather than a red flag — the metric worth tracking there is season-completion (does the viewer finish the season across multiple sessions), not single-session completion. Stand-up specials and movies, by contrast, are single-session formats, so a low completion rate on those would be the actual warning sign.

9.3 Peak Viewing Hours (Simulated)

With play-event timestamps, viewing volume could be aggregated by hour of day to find peak demand windows — useful for CDN capacity planning, release-timing decisions (when to drop a new title), and staffing for live or interactive content. The illustrative pattern below (a typical evening peak roughly **7–10 PM**, highlighted in red) is fabricated for demonstration; the actual peak window would vary by region and time zone in a real deployment.

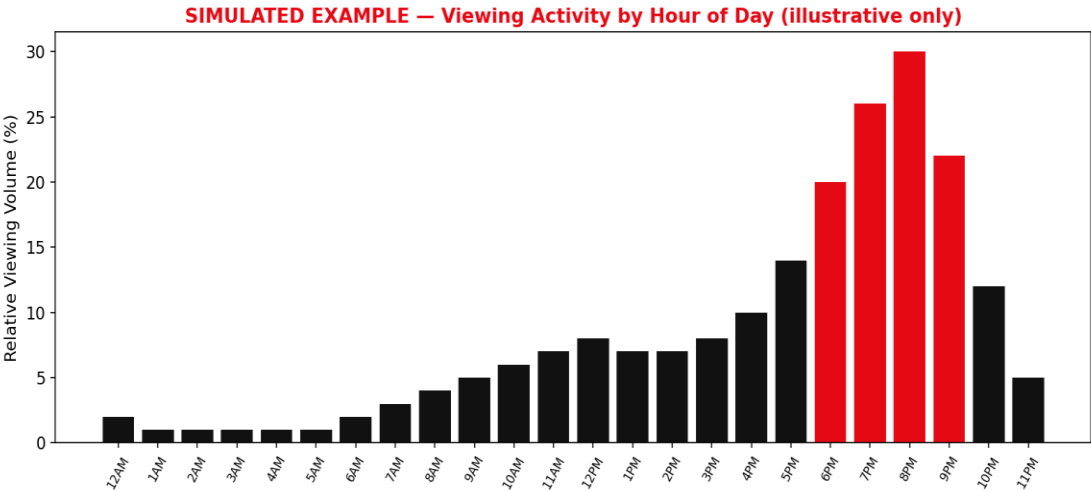


Figure 8 (SIMULATED) — Viewing activity by hour of day — illustrative numbers only

9.4 Weekday vs. Weekend Viewing (Simulated)

Streaming platforms typically report elevated viewing volume on Friday through Sunday. The illustrative pattern below (Saturday as the simulated peak day) follows that general shape and would, with real data, inform when to schedule major title drops and marketing pushes.

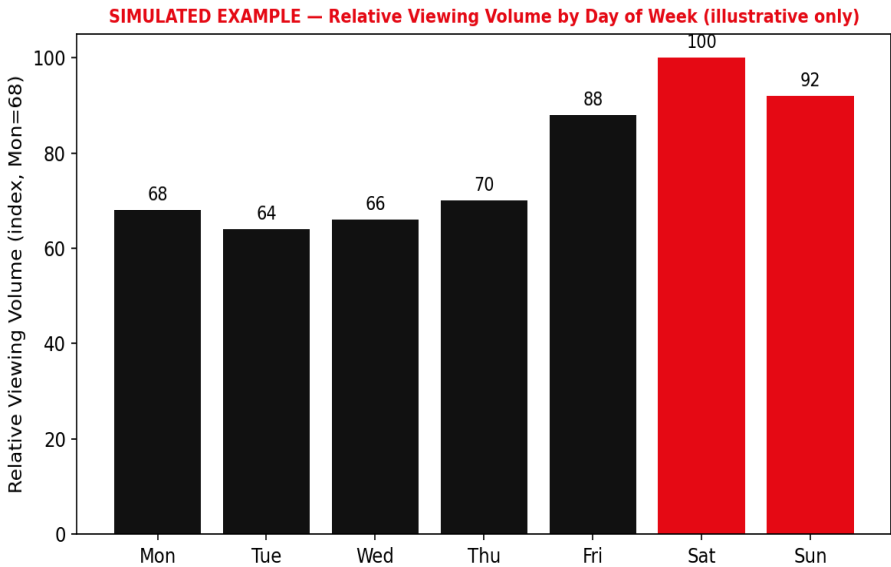


Figure 10 (SIMULATED) — Relative viewing volume by day of week — illustrative numbers only

9.5 Device & Platform Mix (Simulated)

A complete viewing-behavior picture would also break volume down by device (TV, mobile, tablet, desktop/browser) and by connection type, since both affect video-quality decisions and app-design priorities. No device data exists in this catalog export, so this is listed as a placeholder dimension rather than charted:

- Connected TV — typically the majority of total minutes watched on most streaming platforms (illustrative expectation, not measured here)
- Mobile — shorter sessions, more relevant to commute/break-time viewing windows
- Desktop/browser — smallest share on most platforms, more relevant to workplace or shared-computer viewing

9.6 Viewer Preferences (Simulated)

Cross-tabulating simulated view counts against the catalog's existing genre, country, and rating fields (Sections 5–7) would show which combinations over-index with viewers relative to their share of the catalog — e.g., a genre with a small catalog footprint but disproportionately high viewership would flag as under-supplied relative to demand. This cross-tab requires real engagement data to compute and is not run here.

What would be needed to make this section real:

- A play-events or session log (title ID, user ID, timestamp, watch duration, device) joined to the existing title_id key
- User-submitted ratings/thumbs data, if popularity-by-satisfaction (not just volume) is also in scope
- A defined time window (e.g., trailing **90** days) since, unlike the static catalog snapshot, viewing activity is continuous
- Regional/time-zone tagging on timestamps, since peak-hour and weekday patterns are local, not global

10. Summary of Key Metrics

Metric	Value
Total titles in catalog	8,807
Movies / TV Shows	6,131 (70%) / 2,676 (30%)
Countries represented	85
Content categories (primary genre)	36
Peak addition year	2019 (2,016 titles)
Most common rating	TV-MA (36.4% of catalog)
Top content category	Dramas (1,600 titles)
Top sourcing country	United States (3,211 titles)
Average movie runtime	100 min (median 98 min)
Average TV seasons per show	1.8
Titles with complete metadata	60.7%
Release year range	1925–2021

11. Business Insights

Reading the metrics above as business signals rather than raw counts:

Content strategy

- Dramas and Comedies together anchor over a third of the catalog. That concentration is a proven demand base, but it also means the next incremental acquisition dollar is likely to have more differentiating impact in under-represented, growing categories (e.g., Documentaries, International TV Shows) than in adding further volume to the already-dominant genres.
- TV Shows' rising share of yearly additions (Figure 1) suggests acquisition budget is already shifting toward series — consistent with series driving more recurring engagement per title than a one-off film. This shift is worth reinforcing rather than correcting.

Market concentration & regional expansion

- **36%** of the catalog is sourced from the United States alone, with India a distant second at **11%**. That concentration is a single-market dependency risk for a global platform — the other **83** countries in the catalog are long-tailed, which points to local-production investment (rather than further US licensing) as the higher-leverage lever for growing engagement in non-US markets.
- A mature-skewing rating mix (TV-MA + R + NC-**17** together make up over **45%** of the catalog) works well in markets with looser content standards but can cap addressable audience in markets with stricter local ratings norms — a useful cross-check before greenlighting regional expansion.

Catalog risk management

- **67.0%** of TV Shows never made it past a single season. At scale, that's a large share of development spend that did not convert into a renewed, ongoing series — a strong candidate for a renewal-risk model that flags shows early using genre, country, and rating as predictors, before a second season is greenlit.
- **8.2%** of titles are missing two or more of Director, Cast, and Country. Since these fields feed search, recommendations, and browse-by-cast/director experiences, this is a

direct, fixable discoverability gap — and one that's cheap to close relative to acquiring new content.

Timing signal

- Additions peaked in **2019 (2,016 titles)** and declined in the two years after. Because this snapshot ends midway through **2021**, that decline could reflect a genuine slowdown in acquisition pace or simply a partial final year — it should be confirmed against a current catalog feed before being used to justify any budget decision.

Taken together, these four insight groups point toward the same operating principle: descriptive analytics is most valuable when it changes where attention is spent next, not just when it produces an accurate chart. A dashboard that shows genre concentration is descriptive; a decision to redirect the next quarter's acquisition budget toward an under-represented, high-potential genre is where that descriptive work turns into business value.

12. Next Steps to Move From Descriptive → Predictive

This report answers what the catalog looks like today. The natural next layer, if useful:

- Content-gap forecasting by genre/country/rating to guide acquisition and licensing spend
- Renewal-risk modeling for single-season shows, using genre, rating, and country as predictors
- Metadata-completion prioritization, ranking incomplete records by expected discoverability impact
- Joining a real viewing-activity feed to replace the simulated figures in Section 9 with actual popularity, duration, and peak-hour metrics

Happy to build out any of these next, or swap in a live catalog feed (API export, database connection, etc.) if available — the same pipeline and chart structure above would apply directly.

A reasonable sequencing for a real team picking this up would be to tackle the metadata-completion pass first, since it is the lowest-cost, lowest-risk of the four items and improves the input quality that every downstream model would depend on. Renewal-risk modeling and content-gap forecasting would follow once a full viewing-activity feed is available, since both need real engagement data rather than catalog metadata alone to produce anything more useful than a rules-of-thumb heuristic.

13. Appendix A — Data Dictionary

A field-by-field reference for the **12** columns in the underlying netflix_titles.csv export, included so any reader can trace a chart back to the exact source column it was built from.

Field	Description
Type	Whether the row is a Movie or a TV Show
Title	The title of the movie or show
Director	Director credited on the title; blank for 2,631 titles, most often ensemble-credited TV Shows
Cast	Lead cast members listed for the title; truncated with "..." where the source list was long
Country	Primary country (or countries) of production
Date Added	Date the title was added to the Netflix catalog, spanning 2008-01-01 to 2021-09-25
Release Year	Original theatrical or broadcast release year, ranging 1925–2021
Rating	Maturity rating (TV-Y through NC- 17)
Duration	Movie runtime in minutes, or TV Show season count
Genre(s)	One or more content categories per title (36 distinct categories across the catalog)
Description	Short synopsis; truncated with "..." in the source export where long

14. Appendix B — Assumptions & Limitations

- This is a single point-in-time snapshot (through **2021-09-25**), not a live feed — any trend described as "declining" or "growing" reflects the shape of this export only and should be re-checked against a current pull before being used operationally.
- Missing fields are treated as genuinely missing, never imputed or guessed — a blank Director field is reported as blank, not filled with a placeholder or a statistical estimate.
- Truncated free-text fields (Director, Cast, Country, Genre(s), Description) may undercount multi-value entries — a title with many co-stars may show fewer Cast members than actually credited.
- Section 9's viewing-behavior figures are entirely simulated and are clearly labelled as such throughout; they illustrate the shape a real engagement analysis would take, not actual Netflix viewership.
- Country of production is treated as a single primary country per title for charting purposes, even though some titles list multiple co-producing countries in the source data.
- "Best-performing" in Section 5 is defined strictly by title count, not by runtime, engagement, or revenue — a genre with fewer, longer titles could represent more content-hours than its title count alone suggests.

15. References & Data Sources

- Kaggle — "Netflix Movies and TV Shows" dataset (netflix_titles.csv), mid-**2021** catalog snapshot
- Python pandas documentation — data profiling, grouping, and aggregation reference
- Matplotlib documentation — chart construction reference for all figures in this report
- Claude AI (Anthropic) — assisted with data profiling, chart generation, and report drafting throughout this project

16. Acknowledgements

I would like to thank my instructors and mentors at Babu Banarasi Das University for their guidance throughout this project, and the maintainers of the Kaggle Netflix Movies and TV Shows dataset for making a clean, well-documented catalog export publicly available for exploratory analysis.

Thank You

— *Afreen Ansari*